

Tensor networks for neuro-symbolic AI

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- ▶ Tensor networks for neuro-symbolic AI
- ▶ Maximum entropy distributions
- ▶ Summary and Outlook

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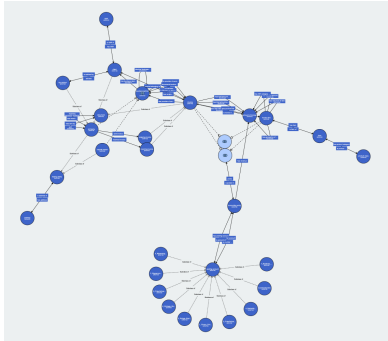
Toy accounting rule system

Assume you want to book one of two accounts A1 and A2 depending on a feature F of an invoice, based on two rules:

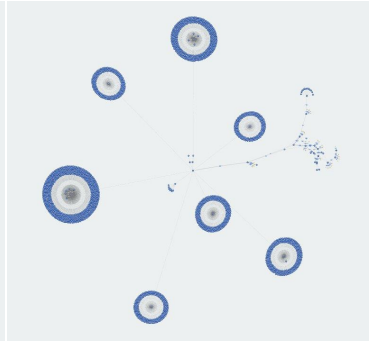
- ▶ Exactly one account must be booked.
 - ▶ If feature F is present on the invoice, the account A1 is typically booked.
-
- ▶ Accounting rules are formalised in propositional logics, triggered by receipts properties.
 - ▶ These rules can either be hard restrictions, or prone to exceptions.

We apply tensor networks for neuro-symbolic use cases like this!

The variables and main relations of the DATEV accounting ontology¹:



Coarse Visualization



Detailed Visualization

¹<https://github.com/EnexaProject/enexa-use-case-1-business-software-solutions>

Tensors: Natural description of factored representations

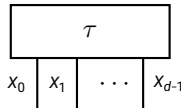
Factored representations are collection of variables $X_{[d]}$, each with a finite set of states $[m_k] = \{0, \dots, m_k - 1\}$ [RN21]. The set of states is then the cartesian product

$$\times_{k \in [d]} [m_k]$$

Knowledge about the system, for example the probability of each state, is a **tensor**

$$\tau [X_{[d]}] : \times_{k \in [d]} [m_k] \rightarrow \mathbb{R}$$

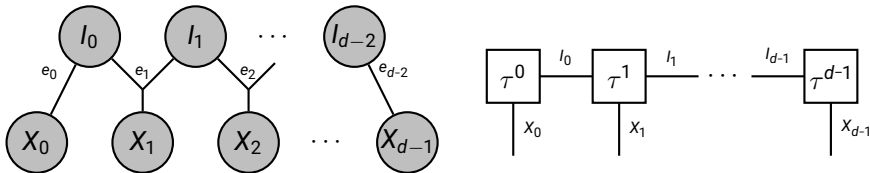
We represent them by diagrams with legs decorated by the variables:



To overcome the curse of dimensionality, tensors are decomposed into networks on hypergraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

$$\tau^{\mathcal{G}} [X_{\mathcal{V}} = x_{\mathcal{V}}] = \prod_{e \in \mathcal{E}} \tau^e [X_e = x_e]$$

Example: TT format



Beyond black-boxes: Hypergraphs encode symbolic structure.

- Probabilistic perspective: Hypergraphs encode **conditional independencies**.
- Neural and logical perspective: Hypergraphs encode **function decompositions**.

Graphical models: Interpretation of tensor networks as probability distributions.

Separation in hypergraphs implies conditional independencies in the distribution.

Neural sparsity

Functions of interest decompose into simple functions (neurons).

Capture by tensor networks: **Basis encodings** of a function

$$f : \times_{k \in [d]} [m_k] \rightarrow \times_{\ell \in [p]} [m_\ell]$$

is the tensor

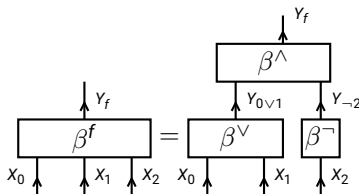
$$\beta^f [Y_{[p]}, X_{[d]}] = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \epsilon_{f(x_{[d]})} [Y_{[p]}] \otimes \epsilon_{x_{[d]}} [X_{[d]}] .$$

Basis calculus: Contraction with one-hot encoded input yields the one-hot encoded output.

Logical syntax translates in tensor network decompositions:

- Basis encodings of the logical connectives $\wedge$, $\vee$, $\neg$ are elementary tensors.

Example: The basis encoding of $f[X_{[3]}] = (X_0 \vee X_1) \wedge \neg X_2$ is decomposed as:

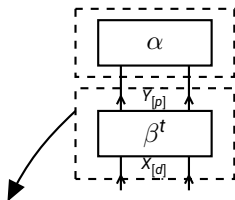


Outlook: First-order logics

Objects enumerated in additional tensor legs.

Computation-Activation Networks (CompActNets)

A novel tensor network architecture captures logical and probabilistic models (see [GSFE26]):



Activation of a statistic by an activation core, for **knowledge bases**

$$\alpha [Y_{[p]}] = \bigotimes_{\ell \in [p]} \begin{pmatrix} 0 \\ 1 \end{pmatrix} [Y_{\ell}]$$

and for **exponential families**

$$\alpha [Y_{[p]}] = \bigotimes_{\ell \in [p]} \begin{pmatrix} 1 \\ \exp [\theta [L = \ell]] \end{pmatrix} [Y_{\ell}]$$

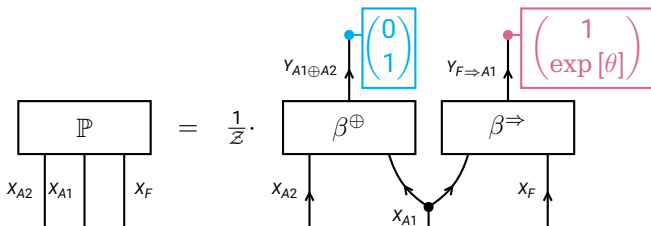
Computation of a statistic

$$t : \times_{k \in [d]} [m_k] \rightarrow \times_{\ell \in [p]} [p_{\ell}]$$

by basis encoding

Toy accounting system as a CompActNet

We represent the toy accounting system as a Computation-Activation Network:



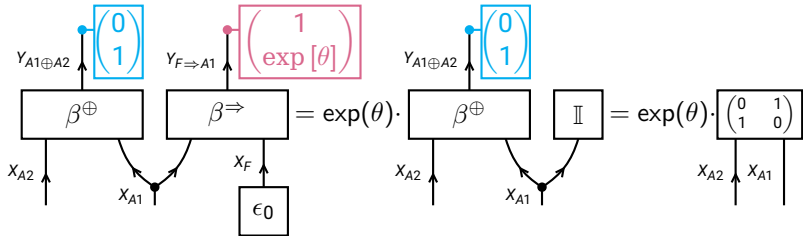
- ▶ The hard activation vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix} [y_{A1 \oplus A2}]$ ensures that the first rule is always true.
- ▶ The soft activation vector $\begin{pmatrix} 1 \\ \exp[\theta] \end{pmatrix} [y_{F \Rightarrow A1}]$ introduces the trainable parameter $\theta \in \mathbb{R}$ controlling the satisfaction probability of the second rule.

Main inference principle

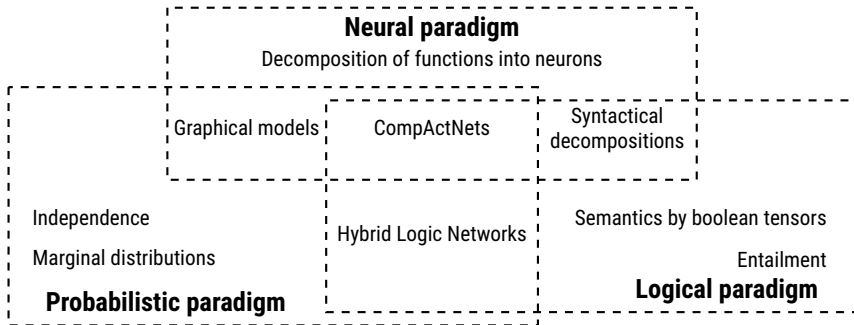
Model inference amounts to tensor network contractions.

- Probabilistic inference: Contractions are marginal probabilities.
- Logical inference: Contractions are model counts, which decide entailment.

Example: Distribution of X_{A1} and X_{A2} conditioned on $X_F = 0$ is parallel to



Tensor networks provide a unifying language to capture the different paradigms of AI [GSFE26]:



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Assume we observe the following satisfaction rates of the toy accounting system:

$$\mu [L = 0] = 1$$

$$\mu [L = 1] = 0.75$$

What distribution should we choose to model the system?

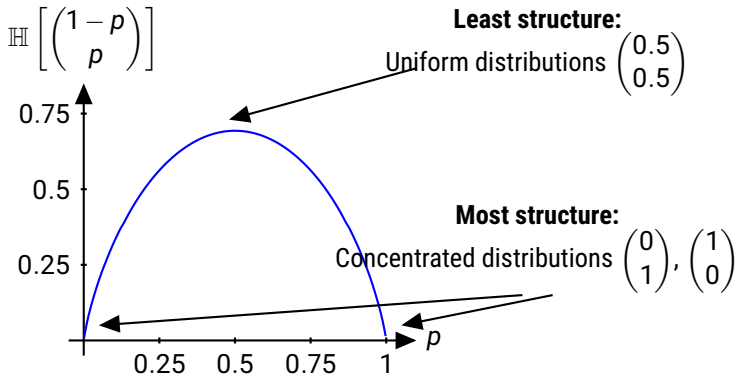
Intuition: Choose the distribution with the least structure, which reproduces the observed satisfaction rates.

Entropy: Quantifying the amount of structure

The entropy of a distribution $\mathbb{P} [X_{[d]}]$ is a measure of its structure:

$$\mathbb{H} [\mathbb{P} [X_{[d]}]] = - \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P} [X_{[d]} = x_{[d]}] \cdot \ln [\mathbb{P} [X_{[d]} = x_{[d]}]]$$

Example: Entropy of Bernoulli distributions



Shannon Entropy: Natural measure of structure (negative information content) of a distribution.

Maximum entropy principle

Among those distributions reproducing certain characteristics of observed data, choose the one with the least structure (=maximum entropy).

Formally the maximum entropy problem is

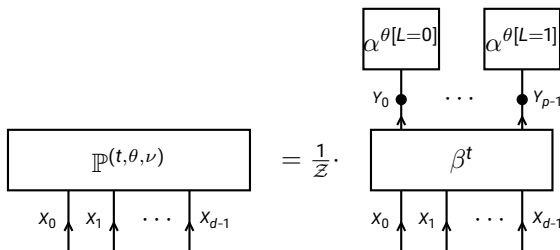
$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}]} \mathbb{H}[\mathbb{P}[X_{[d]}]] \quad \text{s.t.} \quad \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L] \rangle_{[L]} = \mu[L] .$$

Exponential families are maximum entropy distributions

The exponential family of a statistic t is the set of distributions

$$\mathbb{P}^{(t,\theta)} [X_{[d]}] = \frac{1}{\mathcal{Z}(\theta)} \exp \left[\langle \theta, \sigma^t [X_{[d]}, L] \rangle_{[L]} \right]$$

We represent any member as a Computation-Activation Network:



Classical theorem: Any member of the exponential family is a solution of the maximum entropy problem to its mean parameter $\mu [L]$ [KF09].

Realizable mean parameters form a convex polytope, the **mean polytope** of the statistic t :

$$\mathcal{M}_t = \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : X_{[d]} = x_{[d]} \in \prod_{k \in [d]} [m_k] \right)$$

Exponential families can only realize the relative interior of the mean polytope.

What are the solutions to the mean parameters on the boundary of the mean polytope?

Characterization by the face of the mean polytope to which the mean parameter belongs.

We represent faces in activation cores [GE26]

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \sum_{v \in \mathcal{F}} \epsilon_v [Y_{[p]}]$$

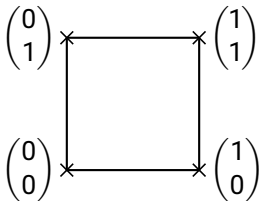
Theorem (Generic maximum entropy representation [GE26])

For any mean parameter on the relative interior of a face $\mathcal{F}$, we find a parameter $\theta [L]$ such that the corresponding maximum entropy is

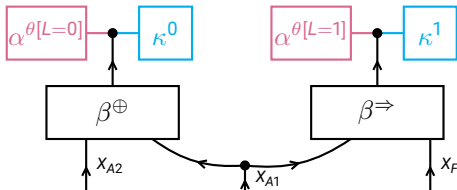
$$\frac{\langle \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}], \alpha^{\theta} [Y_{[p]}] \rangle_{[X_{[d]}]}}{\langle \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}], \alpha^{\theta} [Y_{[p]}] \rangle_{[\emptyset]}}$$

Toy accounting example

The mean polytope is

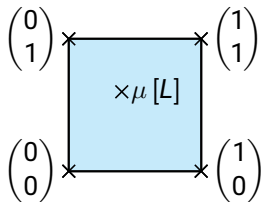


Any solution of the maximum entropy problem is a CompActNet with elementary activation.

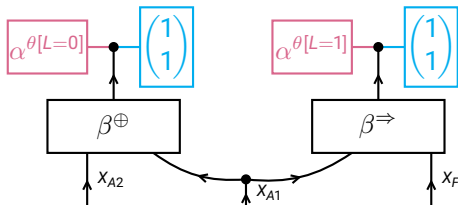


Toy accounting example

The mean polytope is

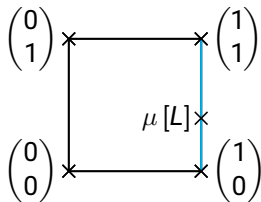


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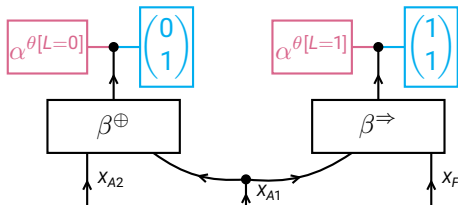


Toy accounting example

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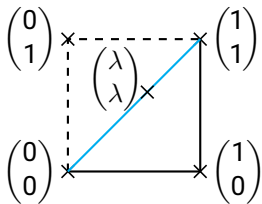
Any solution of the maximum entropy problem is a CompActNet with elementary activation.



Consider the statistic

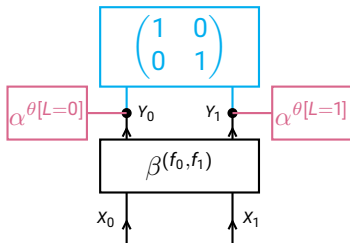
$$f_0[X_0, X_1] = (X_0 \wedge X_1) \quad , \quad f_1[X_0, X_1] = (X_0 \Rightarrow X_1)$$

The mean polytope of this statistic is



The blue face cannot be represented by an elementary tensor.

Representation of the maximum entropy distribution for a mean parameter in the blue face:



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Tensor networks

- ▶ Tensors appear naturally as factored system representations
- ▶ **Sparsity principle**: Decomposition into tensor networks
- ▶ **Inference principle**: Tensor network contractions

Implementations in python packages:

- ▶ `tnreason`²: Tensor network based reasoning
- ▶ `qcreason`³: Quantum inference for contractions

²<https://pypi.org/project/tnreason/>

³<https://pypi.org/project/qcreason/>

Thank you very much for your attention!

- [GE26] Alex Goessmann and Martin Eigel.
Tensor network representations of discrete maximum entropy distributions via mean polytopes, September 2026.
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